

Werner Bisschoff

SOFTWARE ENGINEER

AWS Certified Solutions Architect – Associate (In Progress — Target: 6 Weeks)

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Software engineer with 5+ years of experience bridging hardware and full-stack development. I combine technical depth in C++ and Python with a rigorous, agentic engineering methodology. My focus is on resilient system design and productivity-enhancing AI workflows, governed by security, processes, and specification-driven development.

Skills

Primary : C/C++, Python, FreeRTOS, ESP32/ESP-IDF, NFC APDU (ISO 14443-4)

Secondary : NimBLE, UART/SPI/I2C, TinyFrame, BLE, Hardware-in-the-Loop Testing

Foundational : SQL, Docker, CI/CD, Git, Linux

Other : React, Phoenix, AWS, Pulumi, ERPNext

Experience

Founder and Host

Cape Town

DIVERGENT TABLETOP

Jul 2025

- Founded neurodivergent-focused peer community; manage event operations and WhatsApp-based community platform.
- Applied systems thinking to solve community management challenges through software solutions.

Full-Stack Software Engineer

Cape Town

FARO AFRICA

Aug 2024 – Nov 2025

- Extended ERPNext using Python/JavaScript to improve workflows, pricing logic, and operational reporting (SQL) → reduced manual reporting time and improved data accuracy.
- Built mobile application in Expo, including NFC (ISO 14443-4 APDUs) for e-paper price tags and card operations → enabled real-time price updates with fewer tagging errors.
- Migrated internal mobile Retool workflows to Expo → improved performance and enhanced long-term maintainability.
- Provisioned AWS infrastructure with Pulumi; deployed Inngest and PayloadCMS services.
- Diagnosed and resolved issues in a large existing ERPNext installation.
- Introduced LLM-assisted development workflows → improved debugging speed and code review throughput, enabling faster iteration.

Embedded Software Engineer

Remote

INGENICS DIGITAL GMBH (THROUGH VIVA OUTSOURCING)

Mar 2023 – May 2024

- Designed event-driven FSM for I2C-based embedded system (C++/FreeRTOS) → maintainable architecture, fewer bugs.
- Developed ESP32 applications (C/C++, ESP-IDF) with configurable BLE stack, TinyFrame protocol, and FOTA updates.
- Built Python tooling for serial/BLE communication, hardware mocks, and pytest async workflows → accelerated testing.
- Developed active object within QP Real-Time Embedded Framework (QSPY) to simulate device behaviour.

Software Developer

Cape Town

UMAN TECHNOLOGIES

Mar 2021 – Dec 2022

- Implemented SOME/IP RPC services; optimized performance bottlenecks with perf.
- Built IPC/RPC layers (C++/Python) using Cap'n Proto; led small team using Agile practices.

Junior Lecturer

Potchefstroom

NORTH-WEST UNIVERSITY

Feb 2020 – Dec 2020

- Lectured Python and C++ programming for Introduction to Programming for first year IT students in both remote and in-person settings.

Education

B.Eng. Computer and Electronic Engineering

Potchefstroom

NORTH-WEST UNIVERSITY

2020

Focus on embedded systems, software engineering, and electronic design · NFC payment emulator (Kotlin/Android), STM32 firmware (C), PID motor controller (Arduino) · Data analysis with Python/pandas using linear regression, correlation and machine learning